

IoT Monitoring Solution for Event-Based Water Sample Collection —

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Lessons I learnt



Talk map

10–15 min talk + 10 min Q&A

- 1 min — Why event-based sampling (WFD, passive sampler)
- 4 min — What we built: retrofit LoRaWAN notifier
- 7 min — 4 lessons: coverage · cold boot · errors & unknowns · real-world problems
- 1.5 min — 3 recommendations for OSH
- 10 min — Discussion with you

Why event-based sampling?

The passive water sampler

The IoT extension

Design for years without power

Open-source throughout

Lesson 1: Coverage kills the battery

Lesson 2: Cold boot is a feature

- Every wake = full power-on + fresh OTAA join
- **Never hung.** No LoRa lock-up, no watchdog needed.
- Moved Dresden → Berlin (OHS2026): joined local gateway **autonomously**, zero reconfiguration
- A session-preserving sleep design would have needed a hard reset
- Cost: ~2 s bootloader + Arduino init per wake — irrelevant for event-driven use
- Limitation: LoRaWAN gives **no network time** → RTC drifts, re-sync only via USB

Takeaway

For rare-event devices, **stateless cold boot beats elegant power management.**

Lesson 3: Errors and unknowns

Part	What happened	Fix
SAMD21 POR	Supercaps ramp $< 1 \text{ V/ms}$ → POR never fires → undefined boot, manual reset needed	Add MAX809 supervisor next rev
MCP7940 crystal	Common CC4V-T1A (2.1 fF) doesn't oscillate; needs high motional C + $< 70 \text{ k}\Omega$ ESR — not in datasheet	CM7V-T1A from compatibility list
MCP7940 backup	MFP pin behaviour on VBAT differs from datasheet; undocumented	Tolerated — default behaviour + cold-boot design absorb it
Battery holder	Intermittent spring↔tab contact	Solder bridge + shorten standoff

Lesson 4: Real-world problems

Mechanics & materials > electronics

Conclusion: 3 recommendations

1. **Analyze, test, validate, document, repair** the status quo.
2. **Optimize the MVP** for minimum complexity and maximum robustness.
3. **Minimize the requirements** for involved people.

Open hardware, honestly

All files (KiCad, xyzcad, firmware, server) published — including the failures above — under AGPL-3.0 / CERN-OHL-S-2.0.

Thank you — Discussion

Repo & contact: HTW Dresden, Smart Production Systems + Water Management

<https://github.com/TheTesla/HTW-Water-Sample-Collector>

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Demonstration at COSMO Wissenschaftsforum in Dresden.

References

- [1] S. Helmert, F. Bahrpeyma, T. Haas, and D. Reichelt, "Implicit surface geometry based 3D modeling for open science hardware," in *2026 6th international conference on electrical, computer and energy technologies (ICECET)*, 2026, pp. 1–6. doi: [10.1109/ICECET65726.2026.11633154](https://doi.org/10.1109/ICECET65726.2026.11633154).

Backup: numbers at a glance

- Standby: $<10\text{ }\mu\text{A}$ @ 3.6 V (RTC $<2\text{ }\mu\text{A}$, reed leakage $<5\text{ }\mu\text{A}$)
- Battery: 2.4 Ah Li-SOCl₂ + 2× supercaps
- Keep-alive: RTC alarm 1.5 h
- Retries: 4 retry cycles, then stop (21 mAh/day worst case)
- Boot: ~2 s to first app line (bootloader + core)
- Enclosure: IP67, ether-TPU / EPDM-PP cable, PETG mounts
- Server: Paho MQTT → SMTP/STARTTLS, 4 env vars